Deven Desai

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SUMMARY

Software Engineering Master's student and Stanford Hackathon winner with deep systems optimization expertise, C++ and Python fluency, and experience building high performance information retrieval engines. Demonstrated track record developing agentic execution systems, optimizing Qdrant retrieval algorithms, and engineering parallel data execution pipelines.

PROJECTS

Stanford Codex LLM × Law Hackathon *1st Place Winner*
Lead Software Engineer Oct 2025

- Designed an AI-native context retrieval engine using Python, LangChain, and vector embeddings to extract legal precedent from unstructured court transcripts
- Built low-latency search endpoints in Python that queried millions of legal records in under 200 milliseconds, earning top honors out of 50 competing teams

Low-Latency Analytical Search Core *Personal Systems Project*
Systems Software Engineer Sep

WORK EXPERIENCE

Applied AI Intern, *Parspec, Inc., USA* **May 2026 – Dec 2026**

- Improved product-search relevance over a distributor inventory catalog, raising **recall@3 from 50% to 90%** on a **5,000-query** labeled evaluation set, by replacing keyword-only search with **hybrid** lexical and **semantic retrieval** fused via **reciprocal rank fusion**, re-scored with a **cross-encoder re-ranker**.
- Eliminated manual review of several-hundred-page construction documents for distributor bids, extracting structured product records at **88%** precision across **CSI Divisions 26 and 22**, by building a **RAG** pipeline over written specifications and schedules that grounds every extracted field in its source excerpt and auto-enriches contractor **Quantity Take-Off** line items.
- Automated lighting fixture product substitutions across 1.2M+ indexed datasheets at 85 percent match precision by engineering a retrieval pipeline that resolves free text inputs to anchor products and scores candidates via multi attribute filtering and semantic search.

Full Stack Developer, *Terminal Trends, India* **May 2024 – Dec 2024**

- Developed and optimized RESTful APIs using **Node.js** and **Express** to integrate real-time EV charging station data from multiple vendors (e.g., Tesla, government APIs), reducing server response time by **40%** and improving data reliability across the platform.
- Built interactive route-planning features with **React** and **Redux**, enabling users to locate nearby EV stations, view waiting times, pricing per kWh, and other live metrics along their journey, enhancing user convenience and engagement.

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, Go, TypeScript, Rust, SQL, Scala
- **Backend & Cloud:** Node.js, Express, REST APIs, GraphQL, Microservices, Kubernetes, Docker, AWS
- **Data & AI/ML:** Snowflake, PostgreSQL, Vector Databases (Qdrant), Snowpark, RAG Systems, Agentic Workflows
- **Tools & Frameworks:** React, Git, Linux, CI/CD Pipelines, PyTorch, LangChain, Unit Testing

EDUCATION

San Jose State University **San Jose, CA**
Master of Science in Software Engineering

ACHIEVEMENTS

- **Winner – Davis Wright Tremaine LLP Prize**, *Stanford Codex LLM x Law Hackathon (2025)*
  Led a 4-member team to build a **RAG-based legal QA engine** using Python, GPT-4, and vector-embedding retrieval with **cosine similarity**, enabling accurate extraction of contract clauses and delivering **90%+ correct answers**, outperforming 30+ teams.